

Audit Trail & Data Lineage Framework

Template - Generic / Cross-Industry | Aligned to NIST AI RMF & ISO/IEC 42001

Document Type	Audit Trail & Data Lineage Framework
Template Version	1.0
AI System Name	[System Name]
System Owner	[Role – to be completed]
Document Owner	AI Governance Function
Regulatory Alignment	NIST AI RMF · ISO/IEC 42001 · GDPR (where applicable)
Status	Active Template
Classification	Anonymised – Portfolio Use Only

Every AI system must maintain a complete, tamper-evident audit trail from data ingestion through to output delivery. This is required for regulatory compliance, incident investigation, and governance assurance.

1. Purpose & Objectives

This framework defines the requirements for audit trail logging and data lineage tracking across AI system pipelines. It ensures that any AI output can be traced back to its source data, model version, and processing steps - enabling effective governance, incident investigation, and compliance verification.

1.1 Objectives

- Enable full traceability of AI outputs from input source to final delivery
- Provide an immutable record of AI system activity for audit and compliance purposes
- Support incident investigation by preserving the state of inputs, model, and outputs at the time of any incident
- Demonstrate accountability and transparency to regulators, auditors, and stakeholders
- Detect anomalies, drift, and unauthorised changes through monitoring of logged data

2. Data Lineage Map

Complete the following data lineage map for the AI system. Every stage from data source to output must be documented.

Stage	Stage Name	Description	Input From	Output To	Data Owner
1	Data Ingestion	Raw data received from source systems	[Source system / API / file]	Pre-processing layer	[Team / Role]
2	Pre-processing	Cleaning, normalisation, PII redaction, schema validation	Data ingestion layer	Model input layer	[Team / Role]

Stage	Stage Name	Description	Input From	Output To	Data Owner
3	Model Inference	AI/ML model processes prepared input and generates output	Pre-processing layer	Post-processing layer	[ML Team]
4	Post-processing	Output validation, formatting, confidence scoring, flagging	Model inference layer	Delivery layer	[Team / Role]
5	Output Delivery	Final output delivered to end user, system, or downstream workflow	Post-processing layer	[End destination]	[Team / Role]
6	Storage & Archival	Inputs, outputs, and logs stored per retention policy	All stages	Audit log repository	[Infra / Data Eng]

3. Audit Log Requirements

3.1 Mandatory Log Events

The following events must be logged for every AI system regardless of risk classification:

#	Event	Data to Log	Required For	Retention
1	Data ingestion	Source ID, timestamp, record count, schema version, data hash	All systems	[Per policy]
2	Pre-processing completion	Timestamp, records processed, records rejected, PII flags raised, normalisation version	All systems	[Per policy]
3	Model inference	Model ID, model version, timestamp, input hash, output hash, confidence score (if available), inference duration	All systems	[Per policy]
4	Post-processing / validation	Timestamp, validation rules applied, pass/fail counts, anomaly flags raised	All systems	[Per policy]
5	Output delivery	Timestamp, recipient ID, delivery channel, output ID, delivery status, retry count	All systems	[Per policy]
6	Human review event	Reviewer ID (role), timestamp, review outcome, override decision (if any)	HIGH/CRITICAL	[Per policy]
7	Access event	User/system ID, timestamp, resource accessed, access outcome (granted/denied)	All systems	[Per policy]
8	Model version change	Previous version, new version, change author, change date, validation sign-off reference	All systems	Permanent
9	Incident trigger	Timestamp, triggering condition, severity assigned, notified parties	All systems	Permanent

#	Event	Data to Log	Required For	Retention
10	Data deletion / retention action	Record ID, deletion timestamp, deletion trigger, authorised by	Systems processing personal data	Permanent

3.2 Log Quality Requirements

- Logs must be timestamped in UTC with millisecond precision
- Logs must be tamper-evident - write-once or append-only storage with hash verification
- Logs must be structured (JSON preferred) to enable automated querying and alerting
- Logs must include a correlation ID linking all events for a single inference or workflow run
- Log integrity must be verified periodically - checksums validated on a scheduled basis
- Logs must be stored separately from the AI system they monitor - not on the same instance

3.3 Log Retention Schedule

Log Type	Minimum Retention	Maximum Retention	Notes
Inference logs (inputs, outputs)	12 months	[Per DPA / policy]	Reduce to minimum necessary; anonymise where possible
Access and authentication logs	12 months	7 years	Regulatory and security requirement
Model version and change logs	Lifetime of system	Permanent	Required for audit trail continuity
Incident logs	7 years	Permanent	Legal and regulatory hold
Human review logs	12 months	7 years	Accountability record for high-risk decisions
Personal data processing logs	As per GDPR / DPA	Do not exceed legal basis	DPO to advise; align with DPIA
Data deletion records	Permanent	Permanent	Proof of compliance with retention obligations

4. Model Version Control

All AI systems must maintain version control over models, prompts, configurations, and governance documents. Version changes must be logged and linked to validation sign-off before deployment.

Version	Release Date	Change Summary	Validation Reference	Approved By	Status
v1.0	[Date]	[Describe change]	[Validation report ref]	[Role]	[Active/Retired]
v1.1	[Date]	[Describe change]	[Validation report ref]	[Role]	[Active/Retired]
v1.2	[Date]	[Describe change]	[Validation report ref]	[Role]	[Active/Retired]

5. Data Lineage Traceability Test

Before production deployment and after material system changes, a traceability test must be conducted to verify that any AI output can be traced back to its source.

#	Test Scenario	Expected Result	Status	Notes
1	Select a sample AI output and trace back to source data record	Source record identified; data hash matches; pipeline stages logged	[PASS/FAIL]	
2	Verify model version logged against a specific inference event	Model ID and version present in inference log; matches deployed version	[PASS/FAIL]	
3	Verify access log records who retrieved a specific AI output	Access event logged with user ID, timestamp, and outcome	[PASS/FAIL]	
4	Simulate a data deletion event and verify deletion log created	Deletion log entry created with record ID, timestamp, and authorisation	[PASS/FAIL]	
5	Verify incident log created when anomaly threshold breached	Incident log entry created with trigger condition, severity, and timestamp	[PASS/FAIL]	
6	Verify tamper-evidence: attempt to modify a log entry	Modification detected; integrity check fails; alert triggered	[PASS/FAIL]	

6. Monitoring & Alerting

6.1 Automated Monitoring Requirements

- Audit logs must be monitored in real-time or near-real-time for anomalous patterns
- Alerts must be configured for: inference volume spikes; confidence score drops below threshold; access by unauthorised roles; log gaps or integrity failures; data retention policy breaches
- Monitoring dashboards must be maintained for HIGH and CRITICAL systems, reviewed at least weekly

6.2 Monitoring Review Cadence

Risk Level	Log Review Frequency	Dashboard Review	Integrity Check
CRITICAL	Continuous / Real-time alerting	Daily	Weekly automated + monthly manual
HIGH	Daily automated scan	Weekly	Monthly automated
MEDIUM	Weekly automated scan	Monthly	Quarterly
LOW	Monthly automated scan	Quarterly	Semi-annual

7. Access Controls for Audit Logs

- Audit logs must be read-only for all roles except the designated Log Administrator
- Log Administrator access must be restricted to a named individual with senior sign-off
- All access to audit logs must itself be logged - the audit log is itself auditable
- Audit logs must not be co-located with the systems they monitor
- Quarterly access reviews must be conducted for all roles with access to audit logs

8. Framework Review & Sign-Off

Role	Name	Signature	Date
AI Governance Lead			
Engineering Lead / Architect			
Data Protection Officer			
Security Lead			

9. Version History

Version	Date	Summary of Changes	Author
1.0	[Date]	Initial framework - generic cross-industry Audit Trail & Data Lineage aligned to NIST AI RMF and ISO 42001	[Role]